

- 11 (a) e.g. scattering (in metal)
 non-parallel beam (not just “A closer than B”)
 reflection (from metal)
 diffraction in the metal/lattice
any two

B2 [2]

(b) (i) 1. ratio = $e^{\mu x}$
 $= \exp(0.27 \times 4.0)$
 $= 2.94 \text{ (2.9)}$

C1
 A1 [2]

2. ratio = $\exp(0.27 \times 2.5) \times \exp(3.0 \times 1.5)$
 $= 1.96 \times 90$
 $= 177 \text{ (180)}$

C1
 A1 [2]

(do not penalise unit error more than once)

- (ii) each ratio gives measure of transmission
 ratios (in (i)) very different so good contrast

B1
 B1 [2]